

# Atlas-Driven Adaptive Continuation in Inverse Scattering: Literature Review and Pre-Coding Reading Map

## Executive summary

This review treats the proposed “atlas” in the precise sense set out in the project brief: a characterization, on the **current boundary**, of how each physical frequency contributes to each **normal-displacement shape harmonic**, including not only sensitivity magnitude but also residual-dependent update direction; the longer-term aim is to use that characterization to understand complementarity, redundancy, conflict, and eventually continuation in frequency and shape space. The wavefield input/output-mode axis is treated as a second-stage extension rather than allowed to displace the two-axis frequency–shape question. [filecite turn0file0](#)

**Bottom line:** I did not verify a paper that implements the whole proposed loop—trajectory-dependent  $f \times p$  characterization for a penetrable boundary, retaining signed/residual-dependent information and inter-harmonic coupling, followed by *joint* adaptation of data frequencies and active boundary-shape harmonics. That is not a novelty claim; it is a search result bounded by the literature inspected through **23 September 2026**. What the literature does contain is a set of unusually close pieces that almost tile the proposal:

1. **Direct frequency/shape and mode/shape theory exists.** Ammari, Chow and Zou derive scattering coefficients  $W_{nm}$  as Fourier coefficients of the far field and, for a small normal perturbation of a circular, low-contrast inclusion, derive

$$\Delta W_{nm} \propto J_n(kR) J_m(kR) \hat{h}(n - m),$$

which is almost exactly an input-mode  $\times$  shape-mode  $\times$  output-mode sensitivity relation. But it is a **first-order, circular-background, small-contrast** result, not a theorem for arbitrary finite-deformation Müller transmission scattering. <sup>1</sup>

2. **Multifrequency boundary-shape continuation is well established, but normally prescriptive rather than diagnostic-first.** Borges and Greengard use a band-limited boundary and recursive linearization (RLA), moving from lower to higher frequencies; Sini and Thành similarly obtain a low-frequency approximation and refine it at increasing frequencies. These approaches embody the familiar low-frequency/coarse-shape  $\rightarrow$  high-frequency/fine-shape logic but do not first build a frequency-by-harmonic contribution atlas and let the measured interactions determine the continuation. <sup>2</sup>
3. **The closest precedent for your insistence on update *direction*, rather than sensitivity magnitude, is outside boundary scattering.** Oh and Min explicitly form mono-frequency descent directions and reweight their sum using quantities derived from the current residual/backpropagated wavefield. This is much closer to the “which frequency is helping or fighting this shape direction *now*?” question than a Fisher-information or Jacobian-norm map is. It still does not adapt a boundary-harmonic search space. <sup>3</sup>

4. **The closest precedent for changing the model space along a nonlinear inversion trajectory is Adaptive Spectral Inversion (ASI).** Gleichmann and Grote repeatedly adapt both the dimension and basis of the inverse search space from the previous iterate and prove convergence under an angle condition. It does not construct frequency-resolved parameter effects, but it shows that trajectory-dependent search-space enrichment is an established inverse-problem mechanism, not itself a novel ingredient. <sup>4</sup>

5. **A frequency-by-shape heatmap cannot, in general, be the whole atlas.** A Jacobian-norm heatmap measures local visibility; a signed frequency-wise gradient measures residual alignment; a diagonal Gauss–Newton/Fisher heatmap measures local curvature/information. None records the off-diagonal harmonic cross-talk

$$(H_\ell)_{pq} = \text{Re}\langle J_{\ell,p}, C_\ell^{-1} J_{\ell,q} \rangle,$$

which determines whether two apparently “strong” harmonics are actually redundant or nearly indistinguishable. Fichtner and Trampert’s Hessian-based resolution work and García Neeffes and Hawkins’s SVD sensitivity analysis are especially important antidotes to interpreting diagonal sensitivity plots as resolution or reconstructibility. <sup>5</sup>

6. **Information is additive only with qualifications.** For an objective that is literally a sum of independently whitened per-frequency misfits, gradients and Gauss–Newton/Fisher matrices add before the step is computed. Independently computed Gauss–Newton steps generally do **not** add to the joint Gauss–Newton step. Correlated errors between frequencies introduce cross-frequency blocks and destroy a simple decomposition into  $\sum_\ell H_\ell$ . Oh and Min’s weighted gradient sum, Fichtner’s Hessian formulation, and the sensitivity/SVD designs of Li et al. illustrate three different—and non-equivalent—ways of combining information. <sup>6</sup>

7. **Local identifiability and useful nonlinear progress are distinct.** Fisher information, Jacobian singular values and linearized condition numbers concern a neighborhood of the current model. Sini and Thành explicitly study the low-frequency basin in which the objective has controlled behavior; Borges, Rachh and Greengard find that, for penetrable targets, boundary-only optimization can be less robust than a larger volumetric formulation despite its smaller parameter count; and the new 2026 shape-Taylor work gives a systematic language for higher-order shape effects beyond a first derivative. Together these make it unsafe to equate “large  $J_{\ell,p}$ ” with “good nonlinear update.” <sup>7</sup>

8. **The modal extension has strong local precedents but a major conceptual gap remains.** Maximum-information-state work optimizes *physical input states* using derivatives of a physical scattering matrix and Fisher information; 2026 work extends this to multiple parameters and structured modal excitation. Fourier–Galerkin/Müller papers, by contrast, concern discretization of an *internal boundary integral operator*. I found no result justifying the inference that a sparse or weak structure in an external response derivative  $DT$  permits the corresponding entries of the internal Müller matrix  $A$ , or its Fourier interactions, to be dropped while preserving the shape derivatives needed by inversion. <sup>8</sup>

The strongest research boundary therefore lies **not** in showing that high frequencies reveal finer scales, nor in plotting a Jacobian, nor in using frequency continuation. It lies in determining whether a sufficiently rich, trajectory-dependent decomposition of frequency–shape interactions can reliably predict *complementary and useful nonlinear search directions*, and whether such information can justify joint changes to the active data and shape spaces without throwing away cross-harmonic or modal information that matters to the derivative.

## Start here: what to read before coding

The ordering below is by what is most likely to prevent a conceptual mistake in the **first frequency-shape atlas**, not by citation count or general importance. “Full-text inspected” means I could inspect the relevant full paper text in this search; where only abstract/metadata were reliably accessible, that limitation is stated rather than inventing section numbers.

| Priority | Paper                                                                                                                                                                                                                                                     | Why it belongs before the first atlas                                                                                                                                                                                                                                                      | What to read and extract                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First    | <b>Ammari, H.; Chow, Y. T.; Zou, J. (2016), “Phased and Phaseless Domain Reconstructions in the Inverse Scattering Problem via Scattering Coefficients,” SIAM J. Appl. Math. 76(3), 1000–1030. DOI 10.1137/15M1043959; arXiv:1510.03999.</b> <sup>9</sup> | This is the closest analytical paper to the <i>structure</i> of your proposed atlas. It separates incident angular order, observed angular order, physical $k$ , and a normal boundary perturbation. It shows exactly how dangerous it would be to universalize a circular selection rule. | <b>Section 2</b> , especially Definition 2.1, Theorems 2.2–2.4 and eqs. (2.9)–(2.14); then <b>Section 3</b> sensitivity; then <b>Section 4</b> linearized phased reconstruction/resolution. Extract the distinction between $W_{nm}$ , far-field Fourier coefficients and $\widehat{h}(n - m)$ ; record every hypothesis attached to eq. (2.14). Defer the detailed phaseless machinery in Sections 5–8 unless phaseless GPR becomes relevant. Full text inspected. <sup>10</sup> |
| Second   | <b>Borges, C.; Greengard, L. (2015), “Inverse Obstacle Scattering in Two Dimensions with Multiple Frequency Data and Multiple Angles of Incidence,” SIAM J. Imaging Sci. 8(1), 280–298. DOI 10.1137/140982787; arXiv:1408.5436.</b> <sup>11</sup>         | This is the direct baseline that your proposed atlas should beat conceptually: Fourier/ band-limited boundary regularization plus multifrequency RLA. It tells you exactly what “fixed continuation” means in a close obstacle-shape setting.                                              | Read <b>§3.1</b> for single-frequency Newton linearization and band-limited curve updates, <b>§3.2</b> for RLA, and <b>§4.1</b> for bandwidth selection. Note that their update $p_j(t)$ is explicitly Fourier-band-limited and that the paper treats bandwidth as physical regularization. Extract the assumptions behind increasing frequency and the role of incident-direction coverage. Full text inspected. <sup>12</sup>                                                   |

| Priority | Paper                                                                                                                                                                                                                                                           | Why it belongs before the first atlas                                                                                                                                                                                                                                                       | What to read and extract                                                                                                                                                                                                                                                                                                                                                            |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Third    | Oh, J.-W.; Min, D.-J. (2013), <b>"Weighting technique using backpropagated wavefields incited by deconvolved residuals for frequency-domain elastic full waveform inversion,"</b> <i>Geophys. J. Int.</i> 194(1), 322–347. DOI 10.1093/gji/ggt107. <sup>3</sup> | This is the best close precedent I found for your phrase "strength <b>and direction</b> of the contributions" and for recomputing those contributions as the current model/residual changes. It prevents the review from collapsing your idea into Fisher information.                      | Read the derivation of the mono-frequency gradient and especially <b>"Scaling method using weighting factors," eqs. (14)–(16)</b> . Compare their weighted sum of normalized mono-frequency descent directions with conventional summed gradients/pseudo-Hessian scaling. Extract exactly what is residual-dependent and what is normalized away. Full HTML inspected. <sup>3</sup> |
| Fourth   | Fichtner, A.; Trampert, J. (2011), <b>"Resolution analysis in full waveform inversion,"</b> <i>Geophys. J. Int.</i> 187, 1604–1624. DOI 10.1111/j.1365-246X.2011.05218.x. <sup>13</sup>                                                                         | This is the paper to read before interpreting a colorful sensitivity atlas as "resolution." It makes Hessian structure, direction-dependent resolution, image distortion and parameter trade-offs explicit, and it even derives adaptive-parametrization ideas from resolution information. | Read <b>§3</b> on Hessian parametrization, <b>§4</b> on resolution, <b>§5</b> for the physical interpretation of Fourier-transformed Hessians, and <b>§6</b> for the proposed consequences including adaptive parametrization. Appendix A is useful when relating the discussion to matrix-free Hessian actions. Full HTML inspected. <sup>13</sup>                                 |

| Priority | Paper                                                                                                                                                                                                                                                 | Why it belongs before the first atlas                                                                                                                                                                                                                                                                                                                       | What to read and extract                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fifth    | <p><b>Gleichmann, Y. G.; Grote, M. J. (2023), "Adaptive Spectral Inversion for inverse medium problems," Inverse Problems 39(12). DOI 10.1088/1361-6420/ad01d4; arXiv:2303.05229.</b></p> <p>4</p>                                                    | <p>This is the closest methodological precedent for admitting/ changing model directions along the inversion trajectory. The basis itself depends on the previous reconstruction; dimension and basis are repeatedly adapted; an angle condition links the reduced direction to convergence.</p>                                                            | <p>Read the construction of the adaptive spectral search space, the <b>angle-condition/convergence</b> argument, and the numerical inverse-scattering examples. Extract what information triggers basis/dimension changes and what their convergence condition guarantees—and does <i>not</i> guarantee. The abstract and publication were verified; exact internal section numbering was not reliably exposed by the search interface, so I would not trust an AI-generated section number here. 14</p> |
| Sixth    | <p><b>Borges, C.; Rachh, M.; Greengard, L. (2023), "On the robustness of inverse scattering for penetrable, homogeneous objects with complicated boundary," Inverse Problems 39(3). DOI 10.1088/1361-6420/acb2ec; arXiv:2210.11607.</b></p> <p>15</p> | <p>This is the most important reality check for your intended penetrable-object setting. It compares shape-only and volumetric RLA formulations and reports the counterintuitive result that the larger volumetric search space is more robust. That directly warns against reading local shape sensitivity as a proxy for global nonlinear usefulness.</p> | <p>Read the formulation of the penetrable shape inverse problem, the comparison protocol for shape versus volumetric reconstructions, and the discussion of robustness/ failure modes. The paper/ version and principal conclusions were verified from the primary preprint and journal metadata, but the full PDF did not render reliably enough in this search to give safe section/ equation numbers. 15</p>                                                                                          |

| Priority | Paper                                                                                                                                                                                              | Why it belongs before the first atlas                                                                                                                                                                                                                                                                                                                                                                                               | What to read and extract                                                                                                                                                                                                                                                                                                                                                                           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seventh  | Hettlich, F. (2012), "The domain derivative of time-harmonic electromagnetic waves at interfaces," <i>Math. Methods Appl. Sci.</i> 35(14), 1681–1689. DOI 10.1002/mma.2548. <sup>16</sup>          | Your atlas is meaningful only if "shape harmonic" is defined as a genuine shape direction rather than as an arbitrary coordinate coefficient. Hettlich gives a penetrable-interface domain derivative characterized by a transmission problem and connects to the Kirsch/Hettlich shape-derivative tradition. It is Maxwell rather than your exact 2D scalar Müller model, so it is a conceptual foundation, not a drop-in formula. | Focus on the definition of the domain derivative, dependence on interface deformation, and the transmission problem satisfied by the derivative. Pair it with your own scalar-transmission/Müller differentiation. Defer electromagnetic vector details not needed for the scalar first atlas. Primary abstract/metadata verified; full theorem text was not inspected in this pass. <sup>16</sup> |
| Eighth   | Sirgue, L.; Pratt, R. G. (2004), "Efficient waveform inversion and imaging: A strategy for selecting temporal frequencies," <i>Geophysics</i> 69(1), 231–248. DOI 10.1190/1.1649391. <sup>17</sup> | This is the classic "coverage rather than simply more frequencies" precedent. It derives frequency sampling from redundancy in model-wavenumber coverage created by source–receiver offsets. It is useful precisely because it is <i>not</i> residual-dependent: it marks the boundary between a coverage atlas and your proposed trajectory-dependent update atlas.                                                                | Read the model-wavenumber coverage argument and frequency-selection rule, then the synthetic demonstration. Extract the notion of <b>redundant Fourier coverage</b> and its assumptions. Do not import its seismic spatial-wavenumber mapping literally into boundary Fourier modes. <sup>18</sup>                                                                                                 |

A useful reading order is therefore:

**shape/mode meaning → existing fixed multifrequency baseline → residual-dependent combination → full-matrix resolution → adaptive inverse spaces → penetrable nonlinear robustness → rigorous shape derivative → coverage-based frequency selection.**

That sequence is deliberately different from reading chronologically.

For the **wavefield-mode extension**, three additional papers become essential only after the first  $f \times p$  atlas is conceptually sound. Bouchet, Rotter and Mosk derive maximum-information input states from the derivative of a physical scattering matrix and Fisher information; note the published correction to

one supplementary equation, which did not alter their final Fisher-information result. Verreussel, Seifert and Mosk extend that framework in 2026 to multiparameter Fisher matrices and nuisance parameters. Feldman et al. then use partial scattering matrices and their derivatives to choose structured illuminations that selectively excite informative resonant modes. None of these is a theorem about truncating an internal Müller discretization. <sup>19</sup>

**Modal-extension reading:** Bouchet–Rotter–Mosk (2021), DOI 10.1038/s41567-020-01137-4; Verreussel–Seifert–Mosk (2026), arXiv:2606.01445; Feldman et al. (2026), DOI 10.1002/nap2.70119. <sup>20</sup>

## Literature landscape and closest precedents

The useful literature is better organized by **mechanism** than by application field. Five mechanisms recur: frequency continuation; frequency/model-scale coverage; frequency-resolved update combination; adaptive parameter spaces; and input/output modal information.

### Closest-precedent matrix

| Work / strand                              | What is characterized                                                                                                          | How contributions are combined                                                    | Does it make an adaptive decision?                                        | Does the active model space change?                                         | Wavefield-mode information?                                                     | Depends current geometrical residual?                                                   |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Ammari–Chow–Zou 2016</b> <sup>1</sup>   | Analytical dependence of scattering coefficients/far field on small shape perturbations; explicit $n, m, k, \hat{h}$ structure | Not a multifrequency adaptive combination; linearized reconstruction/conditioning | Measurement strategies are analyzed, especially for phaseless stability   | Shape Fourier modes enter explicitly, but no online active-set continuation | <b>Yes—very strongly:</b> $W_{nm}$ is indexed by incident/output angular orders | Geometrical dependence local about perturbation circle; no inversion residual direction |
| <b>Borges–Greengard 2015</b> <sup>12</sup> | Frequency-dependent nonlinear boundary reconstruction with Fourier-band-limited shape corrections                              | Sequential RLA: solution at one frequency initializes the next                    | <b>Yes, frequency schedule,</b> but prescribed rather than atlas-selected | Band-limited shape space regularizes the update                             | No external input/output modal atlas                                            | Geometrical changes iteratively decomposed of signed frequency contributions            |

| Work / strand                                  | What is characterized                                                            | How contributions are combined                                                                                                      | Does it make an adaptive decision?                                         | Does the active model space change?                                    | Wavefield-mode information?                                              | Depends current geometrical residual?                                                      |
|------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Sini-Thành 2012</b> <sup>21</sup>           | Low-frequency objective landscape, then multifrequency refinement                | Low-frequency reconstruction followed by higher-frequency RLA                                                                       | Frequency continuation                                                     | Implicitly coarse-to-fine reconstruction, not joint harmonic selection | No                                                                       | Current geometrical changes emphasize nonlinear basin, no residual-frequency decomposition |
| <b>Sirgue-Pratt 2004</b> <sup>18</sup>         | Frequency/offset → model-wavenumber coverage and redundancy                      | Selects a sparse set of temporal frequencies to obtain desired coverage                                                             | <b>Yes: frequency selection</b>                                            | No adaptive inverse basis in the proposed sense                        | Illumination/offset geometry enters coverage                             | No residual dependence essential acquisition coverage                                      |
| <b>Oh-Min 2013</b> <sup>3</sup>                | Mono-frequency gradient/descent contributions                                    | Explicit weighted sum of normalized frequency-wise directions, with weights derived from deconvolved residual/backpropagated fields | <b>Yes: dynamically reweights frequencies</b>                              | No                                                                     | Wavefields enter the weighting calculation, not as an active modal basis | <b>Yes: this key residual depends on previous precedents</b>                               |
| <b>Fichtner-Trampert 2011</b> <sup>13</sup>    | Full Hessian/resolution operator, directional resolution and parameter trade-off | Curvature is treated as a full operator, not a diagonal sensitivity plot                                                            | Proposes consequences including adaptive parametrization and misfit design | <b>Yes, conceptually</b>                                               | Not your input/output scattering-mode tensor                             | Local to model at which the Hessian is evaluated                                           |
| <b>Gleichmann-Grote ASI 2023</b> <sup>14</sup> | A reduced spectral model space derived from the current iterate                  | Optimization occurs in repeatedly updated subspaces                                                                                 | <b>Yes</b>                                                                 | <b>Yes: basis and dimension both change</b>                            | No                                                                       | <b>Yes: basis depends on previous reconstruction</b>                                       |

| Work / strand                                     | What is characterized                                                                                  | How contributions are combined                                                        | Does it make an adaptive decision?                                                   | Does the active model space change? | Wavefield-mode information?                                             | Depends current geometrical residual?                        |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Li et al. CSEM 2023</b> <sup>22</sup>          | Frequency-dependent sensitivity matrices; frequency-combination quality via singular-value information | Candidate frequency sets scored by SVD-derived criteria; optimized by genetic search  | <b>Yes: select frequency combinations</b> , largely as experimental/inversion design | No joint active model basis         | No                                                                      | Primarily sensitivity design but rather than signed residual |
| <b>Kwon-Lin 2004</b> <sup>23</sup>                | FRF sensitivity rows at candidate frequencies                                                          | Selects rows that are sensitive and linearly independent                              | <b>Yes: frequency-point selection</b>                                                | No                                  | No                                                                      | Local to structural model; not residual-direction            |
| <b>García Neefjes-Hawkins 2025</b> <sup>24</sup>  | Shape-derivative Jacobian and its SVD; visible geometric perturbation modes                            | Full SVD exposes collective shape directions                                          | Diagnostic only                                                                      | No adaptive inverse-space rule      | Multiple incident fields can be stacked, but not an input/output tensor | Geometric specific derivative residual-dependence update     |
| <b>Bouchet-Rotter-Mosk 2021</b> <sup>25</sup>     | Derivative of physical scattering response and Fisher information for a parameter                      | Forms a Hermitian information operator whose eigenstates give optimum coherent inputs | <b>Yes: chooses illumination state</b>                                               | Parameter space is fixed            | <b>Yes: input-state optimization is central</b>                         | Local parameter sensitivity iterative inverse residual       |
| <b>Verreussel-Seifert-Mosk 2026</b> <sup>26</sup> | Multiparameter Fisher matrix from coherent diffraction/scattering                                      | Scalar criteria on the Fisher matrix; nuisance-parameter handling                     | Optimizes input state for multiparameter precision                                   | Fixed parameter set                 | <b>Yes</b>                                                              | Local information the assumed system                         |

The matrix shows why no one adjacent strand should be declared “the same idea.” The closest papers answer *different pieces* of the question.

## Multifrequency continuation: the established baseline

Recursive linearization predates the current boundary-shape papers and uses increasing frequency to control nonlinearity and resolution. Sini and Thành give a particularly relevant obstacle result: they first seek a rough shape at the lowest frequency, establish quantitative control over a region of the low-frequency objective, then refine at higher frequencies. Borges and Greengard add band-limited

boundary regularization and high-order boundary-integral forward solvers. Bao and Triki provide convergence/error analysis for a general multifrequency inverse-medium RLA under assumptions. <sup>27</sup>

The key difference from the proposed research direction is therefore **not “adaptive versus nonadaptive” in the loose sense**. RLA already adapts the current reconstruction as frequency increases. The distinction is that the next frequency and admissible model bandwidth are largely prescribed by a continuation philosophy; they are not inferred from a measured table or operator describing the *current* interaction of each available frequency with each candidate shape direction. <sup>12</sup>

For penetrable homogeneous objects, Borges, Rachh and Greengard extended the comparison to boundary-shape versus volumetric reconstructions. Their reported result—that the volumetric formulation is more robust despite many more degrees of freedom—is highly relevant to atlas interpretation: restricting the search space to a seemingly “efficient” shape representation does not by itself guarantee a better nonlinear landscape. <sup>15</sup>

The seed literature here was substantially correct. Sini and Thành is “**Inverse acoustic obstacle scattering problems using multifrequency measurements,**” *Inverse Problems and Imaging* 6(4), 749–773 (2012), DOI 10.3934/ipi.2012.6.749. Bao and Triki’s “**Error Estimates for the Recursive Linearization of Inverse Medium Problems,**” *J. Comput. Math.* 28(6), 725–744 (2010), DOI 10.4208/jcm.1003-m0004 is an analysis paper rather than another boundary-shape implementation. <sup>28</sup>

## Frequency–spatial-scale coverage and frequency selection

Sirgue and Pratt is the canonical transferable precedent for an “atlas” built from **coverage** rather than gradients: source–receiver offsets and temporal frequencies cover model wavenumbers redundantly, allowing many frequencies to be omitted while retaining desired coverage. Their argument is particularly useful as a conceptual foil for your project because it shows how a deterministic frequency-to-spatial-scale relationship can drive frequency choice without using a current residual. <sup>18</sup>

Modern diffraction-tomography theory makes this viewpoint especially explicit. Kirisits, Quellmalz and Setterqvist rigorously generalize the Fourier diffraction theorem and analyze Fourier-space coverage as illumination direction, object orientation and frequency change. But this remains a **linearized scattering-potential/Fourier-coverage** statement, not a theorem that boundary normal harmonic  $p$  at a nonlinear current interface is observable whenever a corresponding Euclidean spatial wavenumber lies in a coverage set. <sup>29</sup>

Li et al.’s 2023 land-CSEM paper is a closer precedent to *combining frequency-resolved characterizations*: it computes sensitivity information for frequency candidates, evaluates combinations using singular-value-derived measures and searches for favorable frequency sets. An important result is that simply adding more or more widely separated frequencies need not monotonically improve the chosen resolution metric. Yet it is still much closer to experimental/frequency design from sensitivity matrices than to an online residual-dependent shape continuation. <sup>22</sup>

Kwon and Lin’s structural-model-updating paper makes the redundancy point in still cleaner finite-dimensional form: frequency points correspond to rows of an FRF sensitivity matrix, and desirable points should be both sensitive to the updating parameters **and linearly independent** of already selected information. That principle transfers directly to your concern that a large frequency–harmonic magnitude does not tell you whether a frequency adds anything new. <sup>23</sup>

## Residual-dependent combination and full-matrix resolution

Oh and Min are important because they cross the line from “which frequency is sensitive?” to “which frequency’s current gradient should receive more weight?” Their total descent direction is assembled from mono-frequency directions, and the frequency weights are determined from residual-derived/backpropagated quantities. This is a genuine precedent for **trajectory-dependent frequency contribution weighting**, although the parameter is an elastic medium rather than a boundary shape and no model-harmonic active set is co-adapted. <sup>3</sup>

Fichtner and Trampert, meanwhile, show why the diagonal of such an atlas cannot characterize the way information combines. Their resolution analysis is built around the Hessian and its spatially varying, direction-dependent action; subsequent work with van Leeuwen uses random probing to estimate resolution lengths and **interparameter mappings** without explicitly constructing an enormous Hessian. The lesson transferable to a Fourier boundary basis is that cross-talk is an operator property, not a collection of independent per-parameter sensitivities. <sup>30</sup>

García Neefjes and Hawkins make essentially the same point in a much closer obstacle-shape setting. They discretize the shape derivative, then use the SVD of the Jacobian to find collective geometric modes that most strongly affect the far field. Their results are diagnostic rather than adaptive, and the geometry is star-shaped/sound-soft, but the paper is an excellent warning that the visible directions of the inverse map need not align with your chosen individual Fourier or spline coordinates. <sup>31</sup>

## Adaptive inverse parameter spaces

The Grote line of work is the strongest precedent for model-space enrichment. The 2017 adaptive eigenspace method projects an unknown medium onto a finite-dimensional eigenbasis selected/adapted during inversion; later work extends the framework, and Gleichmann and Grote’s 2023 ASI repeatedly modifies both basis and dimension while incorporating an angle condition into the algorithm. <sup>32</sup>

This precedent matters more to your main question than generic adaptive Fourier–Galerkin PDE solvers. Canuto, Nochetto and Verani’s adaptive Fourier–Galerkin work rigorously adapts Fourier degrees of freedom based on residual marking and coarsening, but its object is the **forward/discretized PDE solution space**, not the inverse shape parameter space or frequency selection. It becomes relevant later, when asking how to reduce a Fourier–Galerkin forward operator while controlling error, not as prior art for the first atlas. <sup>33</sup>

Similarly, Wu, Chen and Ghattas’s offline–online goal-oriented optimal experimental design is useful for notions such as information gain, goal orientation, low-rank structure and separating expensive offline characterization from online design. It does not establish a nonlinear frequency–shape continuation rule. <sup>34</sup>

## Modal scattering and input-state selection

The most important exact modal result is Ammari–Chow–Zou. They define  $W_{nm}$ , show that it is essentially a double Fourier coefficient of the physical far-field pattern, and for a small normal perturbation  $h$  of a circle obtain

$$W_{nm}(B^\delta) - W_{nm}(B) = 2\pi R k^2 \varepsilon_*^{1+\alpha} J_n(kR) J_m(kR) \widehat{h}(n-m) + O(\varepsilon_*^2)$$

under their scaling assumptions. Thus an  $n$ -output/ $m$ -input pair probes the boundary harmonic indexed by their **difference** in that convention. This is much more precise than the informal slogan “input order plus shape order gives output order.” <sup>1</sup>

But the hypotheses are decisive: circular reference shape, infinitesimal normal deformation, low contrast and a particular scattering-coefficient convention. Finite deformations destroy rotational symmetry; general noncircular shapes mix orders; higher-order shape terms create additional coupling paths. The 2026 shape-Taylor papers are relevant precisely because they systematize higher-order shape derivatives for acoustic/electromagnetic scattering, including **transmission** conditions, rather than pretending a first-order selection rule survives unchanged. <sup>35</sup>

Bouchet, Rotter and Mosk supply a different modal ingredient: for a physical scattering matrix depending on a parameter, the input state maximizing parameter information is an eigenstate of an information operator derived from the scattering-matrix derivative. The 2026 multiparameter extension replaces a single scalar information measure with a Fisher matrix and discusses nuisance parameters; Feldman et al. use the same general philosophy to design structured incident fields that selectively address otherwise inaccessible metasurface modes. <sup>36</sup>

Those results justify investigating **input-channel × parameter × output-channel** derivatives. They do **not** justify deleting modes from an internal Müller discretization. Boriskina and collaborators' dielectric-cylinder work verifies that second-kind dielectric boundary integral equations can be discretized with angular/trigonometric Galerkin bases, and later work explicitly compares Müller and elementary BIE formulations near numerical resonances. Cai and Xu prove Fourier-basis matrix compression results for a class of singular boundary integral equations with smooth compact parts. These are forward-solver results; neither gives the missing derivative-preservation theorem connecting internal operator truncation to an external inverse-response sensitivity tensor. <sup>37</sup>

## Timeline of the relevant methodological chain

| Period           | Development and relevance                                                                                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1990s</b>     | Domain/shape differentiation becomes a standard tool in inverse obstacle scattering; Chen's recursive-linearization philosophy establishes frequency continuation as a way of managing nonlinear inverse scattering. Later obstacle/medium papers build directly on that tradition. <sup>38</sup> |
| <b>2003–2004</b> | Ye–Bresler–Moulin derive general Cramér–Rao bounds for parametrically represented boundaries in inverse problems; Sirgue–Pratt formalize frequency selection through model-wavenumber coverage; Kwon–Lin use sensitivity plus row independence to select FRF frequencies. <sup>39</sup>           |
| <b>2007–2010</b> | Bao/Li and Bao/Triki develop regularized RLA and convergence/error analysis for multifrequency inverse-medium problems. <sup>40</sup>                                                                                                                                                             |
| <b>2011–2013</b> | Fichtner–Trampert develop full Hessian resolution analysis; Sini–Thành analyze multifrequency obstacle recovery and the low-frequency landscape; Oh–Min introduce residual-dependent frequency-wise gradient weighting. <sup>41</sup>                                                             |
| <b>2015–2016</b> | Borges–Greengard combine band-limited boundary shape with RLA; Ammari–Chow–Zou derive explicit scattering-coefficient/shape-mode sensitivity and resolution results. <sup>42</sup>                                                                                                                |

| Period           | Development and relevance                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2017–2023</b> | Adaptive eigenspace/spectral inverse methods make the inverse parameter space itself trajectory-dependent; CSEM work explicitly optimizes frequency combinations from sensitivity/SVD information. <sup>43</sup>        |
| <b>2021–2026</b> | Maximum-information-state work turns scattering-matrix derivatives into optimum input-state design; in 2026 this is extended to multiparameter estimation and selective modal excitation. <sup>36</sup>                 |
| <b>2025–2026</b> | Direct obstacle shape-sensitivity SVD studies and high-order shape-Taylor theory sharpen the two weak points of first-order atlases: collective shape modes and breakdown of infinitesimal linearization. <sup>44</sup> |

## What the atlas would actually measure

The phrase “a frequency’s effect on a shape harmonic” denotes several mathematically different objects. Treating them as interchangeable is probably the largest conceptual risk before coding.

Let

$$r_\ell(\Gamma) = F_\ell(\Gamma) - d_\ell,$$

and let the  $p$ -th shape direction be a normal deformation

$$V_p = \phi_p \nu.$$

Your basic Jacobian column is

$$J_{\ell,p} = DF_\ell(\Gamma)[\phi_p \nu].$$

This is already geometry-, acquisition-, contrast-, illumination- and frequency-dependent. In obstacle-scattering shape calculus it is the deformation of the *physical boundary*, not the derivative with respect to whichever Cartesian Fourier coefficients happen to represent the curve. That is the right reason to formulate the atlas in normal-displacement coordinates and only afterwards account for the map from geometric curve coefficients to normal shape directions. Shape/domain derivative theory for scattering interfaces is the relevant foundation here. <sup>45</sup>

## The main candidate meanings of “effect”

| Object                            | Representative expression | Contains current residual? | Contains cross-harmonic information? | What it tells you                                                             | What it cannot tell you                               |
|-----------------------------------|---------------------------|----------------------------|--------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Complex sensitivity column</b> | $J_{\ell,p}$              | No                         | Only if columns are compared jointly | Full first-order data-space response, including phase/direction in data space | Whether the perturbation decreases the present misfit |

| Object                                       | Representative expression                                      | Contains current residual?                                                  | Contains cross-harmonic information? | What it tells you                                                          | What it cannot tell you                                                      |
|----------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Sensitivity magnitude</b>                 | $s_{\ell p} = \ C_{\ell}^{-1/2} J_{\ell, p}\ $                 | No                                                                          | No                                   | Local visibility/SNR after whitening                                       | Sign/direction, redundancy, residual alignment, nonlinear usefulness         |
| <b>Residual-dependent gradient component</b> | $g_{\ell p} = \text{Re}[J_{\ell, p}^* C_{\ell}^{-1} r_{\ell}]$ | <b>Yes</b>                                                                  | Not by itself                        | Whether the current residual projects onto shape direction $p$ , with sign | Correlation with other $p$ 's; curvature; whether a finite step stays useful |
| <b>Gauss-Newton/Fisher block</b>             | $H_{\ell} = \text{Re}(J_{\ell}^* C_{\ell}^{-1} J_{\ell})$      | Usually no for standard additive Gaussian noise, except through $J(\Gamma)$ | <b>Yes</b>                           | Local curvature, correlations, conditioning and identifiability            | Residual sign; higher-order nonlinearity; model discrepancy                  |
| <b>SVD</b>                                   | $J_{\ell} = U_{\ell} \Sigma_{\ell} V_{\ell}^*$                 | No                                                                          | <b>Yes</b>                           | Collective observable shape combinations, null/weak directions             | Which visible direction currently reduces the residual                       |
| <b>Regularized resolution operator</b>       | e.g. $(H + R)^{-1} H$                                          | Depends on formulation                                                      | <b>Yes</b>                           | What a chosen inverse/regularizer would reconstruct or smear               | Global nonlinear basin                                                       |
| <b>Actual GN/Newton step contribution</b>    | solution of a normal equation                                  | <b>Yes</b>                                                                  | <b>Yes</b>                           | Proposed local optimization motion                                         | Not naturally separable by frequency after inversion of the summed curvature |

| Object                         | Representative expression              | Contains current residual? | Contains cross-harmonic information? | What it tells you                                        | What it cannot tell you                                        |
|--------------------------------|----------------------------------------|----------------------------|--------------------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| <b>Finite-step improvement</b> | $\Phi(\Gamma \oplus h) - \Phi(\Gamma)$ | <b>Yes</b>                 | Fully nonlinear                      | Actual local progress for a specified finite deformation | Is expensive and no longer simply a first-order atlas quantity |

The first three rows are closely related but answer different questions. Ammari's shape-sensitivity formulas and García Neefjes–Hawkins's SVD work are fundamentally about the first/local-observability side. Oh–Min's weighting moves into the third row because it uses the current residual. Fichtner–Trampert lives primarily in the Hessian/resolution rows. <sup>46</sup>

## Why whitening and normalization come before interpretation

For an independently whitened least-squares objective

$$\Phi(\Gamma) = \frac{1}{2} \sum_{\ell} r_{\ell}^* C_{\ell}^{-1} r_{\ell},$$

the contribution of frequency  $\ell$  to the first-order gradient in the  $p$ -th coordinate is, up to the chosen complex inner-product convention,

$$g_{\ell p} = \operatorname{Re}(J_{\ell,p}^* C_{\ell}^{-1} r_{\ell}).$$

Without  $C_{\ell}^{-1/2}$ , a frequency can look “important” merely because its receiver noise, source amplitude, calibration scale or physical field amplitude is larger. If instead you normalize each frequency's gradient independently, as some FWI weighting schemes do, that is no longer a neutral display operation: it defines a different effective balance between frequencies. Oh and Min's paper is useful precisely because such scaling is explicit rather than hidden. <sup>3</sup>

The same issue occurs on the **shape** axis. Replacing  $\phi_p$  by  $10\phi_p$  multiplies a Jacobian column and any diagonal information measure correspondingly. On a noncircular interface, “Fourier harmonic  $p$ ” can mean Fourier in the curve parameter, normalized arclength, polar angle, or another transported coordinate; these are not invariantly identical. Your notation  $DF(\Gamma)[\phi_p \nu]$  is therefore conceptually preferable to equating a Cartesian Fourier coefficient of  $(x(t), y(t))$  with a shape harmonic. Shape derivative theory supplies the invariant object; the basis normalization supplies coordinates for it. <sup>47</sup>

## What is genuinely additive across frequencies

If the data errors are independent between frequencies and the total objective is a sum of per-frequency terms,

$$\Phi = \sum_{\ell} w_{\ell} \Phi_{\ell},$$

then differentiation gives

$$g = \sum_{\ell} w_{\ell} g_{\ell}.$$

The Gauss–Newton curvature likewise satisfies

$$H_{\text{GN}} = \sum_{\ell} w_{\ell} \text{Re} (J_{\ell}^* C_{\ell}^{-1} J_{\ell})$$

when the frequency blocks really have block-diagonal covariance. That is the mathematically clean sense in which frequency-resolved **gradients** and **information/curvature matrices** can be superimposed. FWI Hessian work and frequency-wise gradient weighting are compatible with this distinction. 48

If the full residual vector has covariance

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots \\ C_{21} & C_{22} & \cdots \\ \vdots & & \ddots \end{bmatrix}$$

with nonzero cross-frequency blocks, then

$$J^* C^{-1} J$$

contains  $\ell / \ell'$  cross terms. There is no exact decomposition into independent  $H_{\ell}$ 's unless the covariance/weighting structure permits one. This matters for calibration errors, shared source uncertainty and other systematic model discrepancies; describing each frequency independently can otherwise double-count correlated information.

Most importantly,

$$-\left(\sum_{\ell} H_{\ell}\right)^{-1} \left(\sum_{\ell} g_{\ell}\right) \quad / \quad \sum_{\ell} (-H_{\ell}^{-1} g_{\ell}) \quad =$$

in general.

So there are at least three non-equivalent notions of “combining frequency updates”:

sum gradients    /    sum information then solve    /    sum independently computed steps.

Any atlas that stores a “step due to frequency  $\ell$ ” is therefore more algorithm-dependent than an atlas storing  $J_{\ell}$ ,  $g_{\ell}$ , and  $H_{\ell}$ . This is one reason not to choose the final atlas quantity prematurely.

## What a frequency-by-harmonic heatmap necessarily throws away

Suppose the displayed quantity is the diagonal

$$A_{\ell p} = (H_{\ell})_{pp} = \|C_{\ell}^{-1/2} J_{\ell, p}\|^2.$$

Two frequencies can have identical  $A_{\ell p}$  while having completely different off-diagonal structure

$$(H_\ell)_{pq} = \text{Re}\langle C_\ell^{-1/2} J_{\ell,p}, C_\ell^{-1/2} J_{\ell,q} \rangle.$$

One may distinguish  $p$  and  $q$ ; the other may observe almost only their sum. A diagonal heatmap will call both cases equally “strong.” Kwon-Lin’s insistence on linearly independent sensitivity rows, the CSEM use of singular values, Fichtner’s full Hessian resolution analysis and the obstacle-shape SVD study all independently support the same lesson: **information strength without information geometry is incomplete.** <sup>49</sup>

A signed gradient heatmap

$$G_{\ell p} = g_{\ell p}$$

fixes one problem—it tells you whether frequencies currently push a coordinate in the same or opposite directions—but still cannot say whether two harmonics are correlated. Thus “reinforcing versus opposing” has at least two different meanings:

$$g_{\ell,p} g_{\ell',p} > 0$$

means the two frequencies agree on the **coordinate-wise first-order descent direction**, whereas

$$J_{\ell,p}^* J_{\ell',p}$$

or comparisons of their Hessian blocks describe similarity in **data/information geometry**. Those need not agree.

## Why sensitivity is not the same as useful nonlinear progress

Fisher information and Cramér–Rao bounds answer a local statistical question. Bouchet’s maximum-information states likewise optimize local precision with respect to a specified parameter. Ye, Bresler and Moulin’s seed paper should be corrected in this regard: the verified paper is **“Cramér-Rao bounds for parametric shape estimation in inverse problems,” IEEE Trans. Image Process. 12(1), 71–84 (2003), DOI 10.1109/TIP.2002.806249**. It treats parametrically represented boundaries under a general transform/noisy measurement framework, with examples including CT, Fourier imaging and deconvolution; it is **not** specifically a paper on two-dimensional nonlinear wave-scattering shape estimation. <sup>50</sup>

The nonlinear issue is instead whether

$$F(\Gamma \oplus h) = F(\Gamma) + DF(\Gamma)[h] + \mathcal{R}_2(\Gamma, h)$$

has a remainder small enough that the local prediction remains meaningful for the contemplated step. A large Jacobian column can coexist with a poor local model if the current geometry is far from the target or the frequency makes the misfit highly oscillatory. Sini-Thanh’s low-frequency basin analysis and Borges-Rachh-Greengard’s robustness comparison are much closer evidence for this distinction.

<sup>51</sup>

The 2026 papers by Bao, Ma, Lai and Li are therefore unusually timely for this project. **“Shape Taylor Expansion for Wave Scattering Problems,” SIAM J. Appl. Math. 86(2), 482–505, DOI 10.1137/25M1723645**, develops recurrence formulas for higher-order shape derivatives for acoustic and electromagnetic scattering with Dirichlet, Neumann, impedance and transmission conditions. A companion computational paper develops two-dimensional shape Taylor expansions, again including transmission conditions. These papers do not give your adaptive atlas, but they supply a principled

language for asking how long a first-order atlas remains predictive away from the infinitesimal limit.

35

That makes the hierarchy before coding conceptually clear:

visibility /

= identifi

An atlas may contain representations of several levels, but no one scalar quantity silently represents all four.

## Prior-art boundary and unresolved questions

### The frequency-shape atlas

**Already established:** wave scattering has rigorous shape derivatives; Fourier/band-limited boundary updates are standard in inverse obstacle algorithms; low-to-high frequency continuation is mature; linearized frequency and acquisition geometry can be understood in model-wavenumber/coverage terms; sensitivity matrices, Hessians, Fisher matrices and SVDs quantify different aspects of information; and residual-dependent frequency-wise gradient weighting exists in FWI. 52

**Also already established:** the inverse parameter space itself can be modified along the reconstruction trajectory. The adaptive-eigenspace/ASI literature means that “admit new model modes only when appropriate” is not by itself a new methodological principle. 32

**Very close but restrictive:** Ammari–Chow–Zou give an analytical shape-harmonic sensitivity law with explicit physical wavenumber and input/output modal indices. This is the closest mathematical precedent to an atlas cell indexed by  $k$ , shape order and wavefield orders. But because the result is perturbative about a circle and low contrast, it should be treated as a benchmark limiting case for a general numerical atlas, not as a universal selection rule. 1

**What I did not verify:** a penetrable-boundary inversion study that, at each evolving noncircular geometry, computes frequency-resolved normal-harmonic effects, preserves both residual-dependent sign/direction and enough off-diagonal structure to assess complementarity, and then uses that information to decide *both* which physical frequencies and which boundary harmonics should enter the next inverse step. No one ingredient is unprecedented; their coupling under those conditions is the unresolved part found by this review.

The most plausible research questions are consequently narrower and stronger than “can frequencies be mapped to shape scales?”

**Can a local frequency-harmonic characterization predict nonlinear usefulness?** Sensitivity/Fisher information predicts local distinguishability; a gradient predicts first-order objective reduction. Neither guarantees that a finite boundary update helps when the current object is geometrically inaccurate. The new higher-order shape-Taylor theory gives one possible theoretical language for examining this gap, while the robustness literature shows the gap is practically real. 53

**How much of the full per-frequency operator must an atlas retain?** A diagonal  $f \times p$  plot is interpretable but throws away  $p$ -to- $q$  ambiguity. A full  $H_\ell$  retains cross-talk but grows quadratically in shape bandwidth. SVD/eigenspace and random-probing literatures suggest compressed descriptions

are possible, but I did not find a paper establishing what compressed object is sufficient for adaptive boundary-harmonic continuation. <sup>54</sup>

**How should “the same shape harmonic” be transported along a nonlinear trajectory?** On a circle the Fourier modes have a natural global meaning. On a general smooth non-star-shaped curve, a harmonic in an arbitrary parameter is not an invariant geometric object, while a normal deformation basis changes as  $\Gamma$  changes. This is partly a shape-calculus/basis-design issue rather than a frequency-selection problem, and resolving it is necessary before comparing atlas cells between iterations. <sup>47</sup>

**When are different frequencies complementary rather than merely larger in aggregate?** Sensitivity independence, singular values, full Hessian blocks and coverage all give candidate definitions of complementarity. CSEM and FRF design show that frequency count alone is an inadequate measure; FWI gradients show that present residual directions can even conflict. The unresolved boundary-scattering question is how these notions correlate with successful nonlinear shape recovery. <sup>55</sup>

**Does a fixed physical-frequency/shape-bandwidth proportionality leave recoverable information unused?** RLA provides a strong baseline for testing this question, not evidence for the answer. Diffraction/coverage theory predicts broad scale-frequency relationships under linearized assumptions, while Ammari's Bessel-weighted mode formula already shows non-monotone modal weights and zeros can occur in special geometries. A convincing project would have to show something beyond rediscovering “higher  $k$  supports higher  $p$ .” <sup>56</sup>

## The modal extension

Here the prior-art boundary is easier to state.

For a circular perturbative target, **external input/output scattering modes and a shape harmonic are analytically linked**. For arbitrary coherent scatterers, **derivatives of the physical scattering matrix can be used to identify maximally informative incident states**. By 2026 this extends to simultaneous multiple-parameter precision and to structured illumination engineered to access particular resonant modes. <sup>57</sup>

What remains unestablished is the step from those physical statements to a **derivative-aware internal Fourier-Galerkin truncation**. Your internal Müller system has schematic form

$$A(\Gamma, k)x = b,$$

while the physical input-to-output response has some form

$$T(\Gamma, k) = \mathcal{O}(\Gamma, k) A(\Gamma, k)^{-1} \mathcal{B}(\Gamma, k)$$

after discretization. Differentiating schematically gives terms such as

$$DT[h] \supset -\mathcal{O} A^{-1}(DA[h])A^{-1}\mathcal{B}$$

plus derivatives of observation/excitation operators.

Therefore an entry of  $A$  that contributes little to the **forward field** can still influence  $DT[h]$  through inverse-operator amplification or cancellation. Conversely, a simple selection rule in  $DT$  can result from cancellations that are invisible at the internal-matrix level. This is why external  $T$ -matrix sparsity, input/output observability and internal Müller compression must remain distinct concepts. The verified

Galerkin/compression literature establishes accurate forward discretizations and matrix compression, not this inverse-derivative equivalence. 58

A substantive modal-extension question would therefore be something like: **under what conditions can modal truncation of the internal forward representation be certified to preserve a specified family of shape derivatives or inverse information quantities?** I did not verify such a result in the searched literature.

## What would make the project routine rather than substantive

Several versions of the project would sit squarely inside established territory.

A study that merely plots  $\|J_{\ell,p}\|$  and observes that increasing frequency makes finer boundary features visible would reproduce familiar resolution/shape-sensitivity ideas. Ammari's analytical work, diffraction coverage and recent obstacle SVD diagnostics already establish close versions of that phenomenon. 59

A scheme that simply increases boundary Fourier bandwidth monotonically with  $k$  while marching from low to high frequency would be principally a variant of established RLA/band-limited obstacle reconstruction. 2

A frequency selector using only static Fisher information or singular values at a nominal model would be close to established experimental-design, CSEM/FRF selection and maximum-information-state methodology. 60

A residual-based scalar reweighting of mono-frequency gradients without shape-space adaptation would sit near Oh and Min and related FWI frequency weighting. 3

And a Fourier–Galerkin Müller truncation justified only by forward-solver matrix decay would be a fast-BIE/model-reduction project rather than a new result about inverse continuation. 61

The project becomes more distinctive only if the **characterization itself** reveals interaction information that fixed continuation or existing scalar information criteria do not capture, and if that information is demonstrably connected to nonlinear inverse behavior rather than only to a prettier sensitivity visualization.

## Extended annotated bibliography and search boundary

### Read next after the essential set

Sini, M.; Thành, N. T. (2012), “Inverse acoustic obstacle scattering problems using multifrequency measurements,” *Inverse Problems and Imaging* 6(4), 749–773, DOI 10.3934/ipi.2012.6.749. Read for the explicit low-frequency basin argument and the transition to recursive linearization. It is an important answer to the question “why should low frequencies help nonlinearity?” rather than merely “what resolution do they have?” 21

Bao, G.; Triki, F. (2010), “Error Estimates for the Recursive Linearization of Inverse Medium Problems,” *J. Comput. Math.* 28(6), 725–744, DOI 10.4208/jcm.1003-m0004. Read for the theoretical assumptions under which multifrequency RLA is convergent. It provides the analytic backdrop against which a nonmonotone/adaptive frequency schedule would eventually need to be compared. 62

**Bao, G.; Li, P. (2007), "Numerical solution of inverse scattering for near-field optics," Optics Letters 32, 1465–1467, DOI 10.1364/OL.32.001465.** This is a concise inverse-medium frequency-continuation precedent: low-frequency Born initialization followed by wavenumber continuation with forward/adjoint solves. 63

**Li et al. (2023), "Effect of frequency combination on the resolution of 3D land-based CSEM and its optimal design," Journal of Applied Geophysics 219, 105251, DOI 10.1016/j.jappgeo.2023.105251.** Read for a concrete sensitivity-matrix  $\rightarrow$  SVD metric  $\rightarrow$  frequency-combination optimization chain. Its main transferable value is how it treats redundancy/complementarity; its limitation is that this is not signed nonlinear update information. 22

**Kwon, K.-S.; Lin, R.-M. (2004), "Frequency selection method for FRF-based model updating," Journal of Sound and Vibration 278, 285–306, DOI 10.1016/j.jsv.2003.10.003.** Particularly useful for the simple principle "sensitive and independent," expressed directly in rows of a sensitivity matrix. 23

**García Neefjes, E.; Hawkins, S. C. (2025), "Far-Field Sensitivity to Local Boundary Perturbations in 2D Wave Scattering," arXiv:2503.09998.** The paper discretizes a shape derivative for star-shaped sound-soft obstacles and applies SVD to identify visible geometric modes. Equation (15) and the associated singular-vector figures are the material to study. It is a close **diagnostic** precedent, not an adaptive inversion. 31

**Fichtner, A.; van Leeuwen, T. (2015), "Resolution analysis by random probing," JGR Solid Earth 120, 5549–5573, DOI 10.1002/2015JB012106.** This becomes useful if full  $H_\ell$  matrices are too expensive. Their stochastic probing estimates resolution measures and interparameter mappings via operator actions. It suggests a way to think about scalable characterization without reducing everything to a diagonal. 64

**Kirisits, C.; Quellmalz, M.; Setterqvist, E., "Generalized Fourier Diffraction Theorem and Filtered Backpropagation for Tomographic Reconstruction," DOI 10.1137/24M167370X.** Read as the clean modern account of how frequency, incidence and observation geometry create Fourier coverage under linearized diffraction tomography. It gives a rigorous reference point for deciding which parts of a frequency-scale atlas are simply Fourier-diffraction physics and which arise from nonlinear boundary geometry. 29

**Wu, K.; Chen, P.; Ghattas, O., "An Offline-Online Decomposition Method for Efficient Linear Bayesian Goal-Oriented Optimal Experimental Design: Application to Optimal Sensor Placement."** Read later for offline/online separation, low-rank information structure and goal-oriented design—not as direct prior art for residual-driven continuation. 34

**Ye, J. C.; Bresler, Y.; Moulin, P. (2003), "Cramér-Rao bounds for parametric shape estimation in inverse problems," IEEE Trans. Image Process. 12(1), 71–84, DOI 10.1109/TIP.2002.806249.** This seed should be **retained but relabeled**: it is a general parametric-boundary CRB paper, not a direct nonlinear inverse-scattering shape paper. Read it for parameter-space Fisher geometry and parameterization dependence. 50

**Bao, G.; Ma, H.; Lai, J.; Li, J. (2026), "Shape Taylor Expansion for Wave Scattering Problems," SIAM J. Appl. Math. 86(2), 482–505, DOI 10.1137/25M1723645, and the 2026 computational companion "Computation of Shape Taylor Expansions for Scattering Problems in Two Dimensions," DOI 10.1137/25M1746999.** These are the most important new papers, through the search cutoff, for investigating the validity horizon of a first-order atlas; both include transmission-type scattering. 35

## Modal-extension foundations

**Ammari, Chow and Zou's scattering-coefficient papers.** Besides the 2016 paper above, “**Super-resolution in imaging high contrast targets from the perspective of scattering coefficients**,” later published in *Journal de Mathématiques Pures et Appliquées*, DOI 10.1016/j.matpur.2017.09.008, is useful for understanding resonant/high-contrast behavior of scattering coefficients. This matters because a naïve atlas can mistake a resonantly large sensitivity for uniformly reliable geometric information. <sup>65</sup>

**Bouchet, D.; Rotter, S.; Mosk, A. P. (2021), “Maximum information states for coherent scattering measurements,”** *Nature Physics* **17**, 564–568, DOI 10.1038/s41567-020-01137-4. Essential for the distinction between “which input mode excites the system strongly?” and “which input state maximizes information about a chosen parameter?” A correction changed a supplementary intermediate equation but not the final Fisher-information result. <sup>66</sup>

**Verreussel, B.; Seifert, J.; Mosk, A. P. (2026), “Multiparameter Maximum Information States for Coherent Diffraction Measurements,”** arXiv:2606.01445.

This recent preprint is directly relevant once multiple shape coefficients are treated simultaneously: Fisher information becomes a matrix, scalar optimality criteria must be chosen, and nuisance parameters can degrade precision. As of the search cutoff it is a preprint, not a peer-reviewed journal result. <sup>26</sup>

**Feldman, N. et al. (2026), “Wavefront Selective Modal Excitations for Optimally Informative Sensing in Fano-Resonant Metasurfaces,”** *Nanophotonics* **15**(9), e70119, DOI 10.1002/nap2.70119.

This verifies the recent lead from the brief. The paper constructs optimally informative structured excitations using a partial scattering matrix and its perturbation derivative and shows that selected inputs can access otherwise poorly excited modes; the reported optimum can substantially outperform conventional excitation in Fisher information. It is sensing/metasurfaces, not boundary inversion. <sup>67</sup>

**Ge et al. (2013), “Multimode Coupling by Boundary Wave Scattering,”** arXiv:1305.5917.

Useful for perturbative mode-coupling intuition in quasi-circular resonators: boundary harmonics mediate coupling between modal orders and higher-order scattering paths matter. It is an analogy to modal coupling, not an inverse-shape theorem. <sup>68</sup>

**Boriskina, Sewell, Benson and Nosich (2004), “Accurate simulation of two-dimensional optical microcavities with uniquely solvable boundary integral equations and trigonometric Galerkin discretization,”** *JOSA A* **21**(3), 393–402, DOI 10.1364/JOSAA.21.000393.

This verifies the dielectric/trigonometric-Galerkin part of the seed. The later Boriskina-Boriskina preprint explicitly compares elementary and Müller dielectric BIE formulations near numerical resonances. These belong in the solver foundation, not in the evidence that an inverse atlas exists. <sup>69</sup>

**Cai, H.; Xu, Y. (2008), “A Fast Fourier–Galerkin Method for Solving Singular Boundary Integral Equations,”** *SIAM J. Numer. Anal.* **46**(4), 1965–1984, DOI 10.1137/070703478.

They exploit decay/compressibility of smooth compact parts of integral operators in a Fourier basis and prove convergence/complexity results. This is useful when considering modal cost reduction, but the theorem concerns forward BIE accuracy, not preservation of shape derivatives or inverse information. <sup>70</sup>

The seed described as “**Jiang–Wang–Yu on Fourier–Galerkin matrix compression**” was not verified as stated. The closest bibliographic hit in this search was Ying Jiang, Bo Wang and Yuesheng Xu on a fast Fourier–Galerkin method for a **biharmonic** boundary integral equation, not the claimed scattering/Müller result. I would therefore not build any modal-truncation claim on that seed without a corrected citation. <sup>71</sup>

## Supporting/background literature

Canuto, Nochetto and Verani's **Adaptive Fourier–Galerkin Methods** supplies rigorous vocabulary for residual marking, enrichment, coarsening and convergence in Fourier spaces, but it is primarily numerical-PDE adaptivity. It should be read only when the question changes from “which inverse shape modes are informative?” to “how should a Fourier trial space be adapted while retaining controlled discretization error?” <sup>33</sup>

Grote, Kray and Nahum's **adaptive eigenspace method for inverse scattering problems in the frequency domain** and the later adaptive-spectral-decomposition work form the immediate ancestry of ASI. They establish that reduced inverse spaces can act as regularization and be modified during the inversion. <sup>72</sup>

Baffet, Gleichmann and Grote's error-estimate work is useful if the eventual project needs quantitative criteria for how accurately an adaptive spectral representation approximates a target; it is secondary for the first atlas because it does not resolve the frequency-versus-shape interaction question. <sup>73</sup>

The 2026 **Besov-space-regularized waveform inversion via frequency marching** is evidence that frequency-marching regularization remains an active topic, but it does not change the prior-art boundary above: it still uses frequency marching to expose progressively finer information rather than first learning a frequency–model-mode interaction atlas. <sup>74</sup>

## Citation chains worth following

The first chain is the **recursive-linearization chain**:

Chen  $\rightarrow$  Sini–Thành / Bao–Triki  $\rightarrow$  Borges–Greengard  $\rightarrow$  Borges–Rachh–Greengard.

It tracks the transition from frequency continuation as a general nonlinear strategy to modern band-limited obstacle and penetrable-object reconstruction. <sup>75</sup>

The second is the **resolution/operator chain**:

Hessian kernels  $\rightarrow$  resolution analysis  $\rightarrow$  random probing / interparameter mappings.

This is the chain to follow before deciding how much off-diagonal information the atlas must preserve. <sup>76</sup>

The third is the **adaptive inverse-space chain**:

adaptive eigenspace  $\rightarrow$  adaptive spectral decompositions  $\rightarrow$  ASI with angle condition.

It is the closest methodological ancestry for selectively admitting new model directions. <sup>77</sup>

The fourth is the **modal-information chain**:

scattering coefficients / shape perturbations + maximum-information states  $\rightarrow$  multiparameter MIS and

This is where the later input-mode  $\times$  shape-mode  $\times$  output-mode project should begin. <sup>57</sup>

## Search vocabulary that proved most discriminating

The generic phrase *adaptive frequency inversion* was much less useful than mechanism-specific terminology. The vocabulary most worth carrying forward is:

| Question                                             | Productive literature terminology                                                                                                 |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Frequency $\leftrightarrow$ spatial/<br>model scales | <b>model-wavenumber coverage</b> , Fourier diffraction theorem, wavenumber illumination, scattering-angle coverage                |
| Boundary mode visibility                             | <b>shape derivative</b> , domain derivative, scattering coefficients, geometric sensitivity modes, Jacobian SVD                   |
| Redundancy/cross-talk                                | <b>Hessian kernel</b> , resolution operator/matrix, interparameter mapping, sensitivity-row independence, singular-value spectrum |
| Residual-aware<br>contribution                       | <b>mono-frequency gradient</b> , frequency-wise descent direction, backpropagated residual, gradient weighting/alignment          |
| Changing inverse space                               | <b>adaptive eigenspace</b> , adaptive spectral inversion, parameter-space enrichment, adaptive parametrization                    |
| Selecting<br>measurements/inputs                     | optimal experimental design, <b>maximum information states</b> , Fisher-information operator, partial scattering matrix           |
| Beyond linear shape<br>sensitivity                   | <b>shape Taylor expansion</b> , higher-order shape derivative                                                                     |
| Modal forward<br>representation                      | trigonometric Galerkin, Fourier–Galerkin compression, Müller boundary integral equation                                           |

These terms map to distinct literatures and should remain distinct rather than collapsed into “adaptive spectral inversion.” <sup>78</sup>

## Search limitations and evidence corrections

The search cutoff is **23 September 2026**. Priority was given to journal versions, original papers, arXiv author manuscripts and institutional repositories. Recent 2026 work was checked explicitly rather than inferred from older bibliographies.

Several seeds require qualification.

The Borges–Rachh–Greengard preprint **is** published in *Inverse Problems* and is directly relevant to penetrable homogeneous targets. <sup>15</sup>

The two recent maximum-information leads are real: the multiparameter paper is an arXiv preprint from 2026, while the Fano-resonant metasurface paper is a 2026 peer-reviewed *Nanophotonics* article. <sup>79</sup>

The Ye–Bresler–Moulin lead was overstated in the brief: the verified paper is a general CRB treatment of parametric shape estimation in inverse problems, not specifically nonlinear two-dimensional scattering. <sup>50</sup>

The Boriskina and Cai–Xu solver leads are real but support **forward Fourier/trigonometric boundary-integral methodology**, not an inverse modal-atlas mechanism. <sup>80</sup>

The claimed Jiang–Wang–Yu scattering-compression lead was **not verified under that bibliographic description**; it should remain excluded from technical conclusions until the citation is corrected. <sup>71</sup>

I also did **not** verify a theorem transferring an angular-momentum selection rule for an external scattering-response matrix to sparsity of an internal Müller matrix, nor a derivative-aware Fourier–Galerkin truncation theorem guaranteeing preservation of  $DF(\Gamma)[h]$ . That absence is an important search result, but not proof that no such paper exists.

The reading programme therefore points to three things that should be understood before any atlas data are interpreted: **what mathematical object each cell represents; what information is hidden off the diagonal; and how far a first-order characterization remains predictive along a nonlinear trajectory**. The literature is already strong on each ingredient separately. The genuinely open-looking territory is the disciplined combination of those ingredients in a changing penetrable-boundary reconstruction, not the individual ideas of frequency continuation, spectral shape parametrization, Fisher information, or modal scattering on their own.

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<sup>1</sup> <sup>10</sup> <sup>46</sup> <sup>57</sup> <sup>59</sup> <https://arxiv.org/pdf/1510.03999>  
<https://arxiv.org/pdf/1510.03999>

<sup>2</sup> <sup>11</sup> <sup>42</sup> <sup>56</sup> <https://epubs.siam.org/doi/10.1137/140982787>  
<https://epubs.siam.org/doi/10.1137/140982787>

<sup>3</sup> <sup>6</sup> <https://academic.oup.com/gji/article/194/1/322/647696>  
<https://academic.oup.com/gji/article/194/1/322/647696>

<sup>4</sup> <sup>14</sup> <https://arxiv.org/abs/2303.05229>  
<https://arxiv.org/abs/2303.05229>

<sup>5</sup> <sup>13</sup> <sup>30</sup> <sup>41</sup> <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-246X.2011.05218.x>  
<https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-246X.2011.05218.x>

<sup>7</sup> <sup>21</sup> <sup>27</sup> <sup>51</sup> <https://www.researchwithrowan.com/en/publications/inverse-acoustic-obstacle-scattering-problems-using-multifrequenc/>  
<https://www.researchwithrowan.com/en/publications/inverse-acoustic-obstacle-scattering-problems-using-multifrequenc/>

<sup>8</sup> <sup>19</sup> <sup>20</sup> <sup>25</sup> <sup>36</sup> <sup>66</sup> <https://www.nature.com/articles/s41567-020-01137-4>  
<https://www.nature.com/articles/s41567-020-01137-4>

<sup>9</sup> [https://portal.mardi4nfdi.de/wiki/](https://portal.mardi4nfdi.de/wiki/Phased_and_phaseless_domain_reconstructions_in_the_inverse_scattering_problem_via_scattering_coefficients)  
**Phased\_and\_phaseless\_domain\_reconstructions\_in\_the\_inverse\_scattering\_problem\_via\_scattering\_coefficients**  
[https://portal.mardi4nfdi.de/wiki/](https://portal.mardi4nfdi.de/wiki/Phased_and_phaseless_domain_reconstructions_in_the_inverse_scattering_problem_via_scattering_coefficients)  
**Phased\_and\_phaseless\_domain\_reconstructions\_in\_the\_inverse\_scattering\_problem\_via\_scattering\_coefficients**

<sup>12</sup> <sup>38</sup> <sup>75</sup> <https://arxiv.org/pdf/1408.5436>  
<https://arxiv.org/pdf/1408.5436>

<sup>15</sup> <https://arxiv.org/abs/2210.11607>  
<https://arxiv.org/abs/2210.11607>

<sup>16</sup> <sup>45</sup> <sup>47</sup> <sup>52</sup> <https://onlinelibrary.wiley.com/doi/full/10.1002/mma.2548>  
<https://onlinelibrary.wiley.com/doi/full/10.1002/mma.2548>

17 <https://cir.nii.ac.jp/crid/1361418518420495360>  
<https://cir.nii.ac.jp/crid/1361418518420495360>

18 78 <https://www.scinovex.com/articles/W2166525224>  
<https://www.scinovex.com/articles/W2166525224>

22 55 60 <https://www.sciencedirect.com/science/article/abs/pii/S0926985123003294>  
<https://www.sciencedirect.com/science/article/abs/pii/S0926985123003294>

23 49 <https://www.sciencedirect.com/science/article/abs/pii/S0022460X03011519>  
<https://www.sciencedirect.com/science/article/abs/pii/S0022460X03011519>

24 54 <https://arxiv.org/pdf/2503.09998>  
<https://arxiv.org/pdf/2503.09998>

26 79 <https://arxiv.org/abs/2606.01445>  
<https://arxiv.org/abs/2606.01445>

28 <https://www.aims sciences.org/article/doi/10.3934/ipi.2012.6.749>  
<https://www.aims sciences.org/article/doi/10.3934/ipi.2012.6.749>

29 <https://epubs.siam.org/doi/10.1137/24M167370X>  
<https://epubs.siam.org/doi/10.1137/24M167370X>

31 44 <https://arxiv.org/abs/2503.09998>  
<https://arxiv.org/abs/2503.09998>

32 43 72 77 <https://edoc.unibas.ch/entities/publication/71572812-72d3-4d89-88fc-5b1d108aed5c>  
<https://edoc.unibas.ch/entities/publication/71572812-72d3-4d89-88fc-5b1d108aed5c>

33 <https://arxiv.org/abs/1201.5648>  
<https://arxiv.org/abs/1201.5648>

34 <https://epubs.siam.org/doi/abs/10.1137/21M1466542>  
<https://epubs.siam.org/doi/abs/10.1137/21M1466542>

35 53 <https://epubs.siam.org/doi/10.1137/25M1723645>  
<https://epubs.siam.org/doi/10.1137/25M1723645>

37 69 80 <https://pubmed.ncbi.nlm.nih.gov/15005404/>  
<https://pubmed.ncbi.nlm.nih.gov/15005404/>

39 50 <https://pubmed.ncbi.nlm.nih.gov/18237880/>  
<https://pubmed.ncbi.nlm.nih.gov/18237880/>

40 63 <https://doi.org/10.1364/OL.32.001465>  
<https://doi.org/10.1364/OL.32.001465>

48 76 <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1365-246X.2011.04966.x>  
<https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1365-246X.2011.04966.x>

58 61 70 <https://epubs.siam.org/doi/10.1137/070703478>  
<https://epubs.siam.org/doi/10.1137/070703478>

62 <https://www.global-sci.com/JCM/article/view/12008>  
<https://www.global-sci.com/JCM/article/view/12008>

64 <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2015JB012106>  
<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2015JB012106>

65 <https://arxiv.org/abs/1410.1253>  
<https://arxiv.org/abs/1410.1253>

67 <https://onlinelibrary.wiley.com/doi/10.1002/nap2.70119>

<https://onlinelibrary.wiley.com/doi/10.1002/nap2.70119>

68 <https://arxiv.org/abs/1305.5917>

<https://arxiv.org/abs/1305.5917>

71 <https://epubs.siam.org/doi/10.1137/140955744>

<https://epubs.siam.org/doi/10.1137/140955744>

73 <https://link.springer.com/article/10.1007/s10915-022-02004-5>

<https://link.springer.com/article/10.1007/s10915-022-02004-5>

74 <https://repository.tudelft.nl/record/uuid%3A2afb9f55-d24e-471b-97e1-5c0f3f1a5631>

<https://repository.tudelft.nl/record/uuid%3A2afb9f55-d24e-471b-97e1-5c0f3f1a5631>